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GREATER NOIDA INSTITUTE OF TECHNOLOGY (Engg. Institute)

1st CLASS TEST, SESSION 2023-2024 (ODD SEM.)

DEPTT:-CSE-IoT

YEAR:-B.TECH-IIND (3rd SEMESTER)

SUBJECT NAME :-DISCRETE STRUCTURE & THEORY OF LOGIC (SUBJECT CODE- BCS-303)

MAX MARKS: 30

TIME: 2 Hours

Note: Answer all questions. Draw neat sketch, wherever necessary. Suitably assume missing data, if any.

SECTION A

Question 1: Attempt all parts in brief.

[1X5=5]

a.	Write the set of all natural numbers x such that $4x + 5 < 60$ in roster form	1	[CO1], K_3 , K_4
b.	Let R be a relation on the set of natural numbers N as $R = \{ (x, y) : x, y \in N, 3x + y = 19 \}$. Find the domain and range of R .	1	[CO1], K_3 , K_4
c.	Draw the Hasse diagram of D_{30} .	1	[CO1], K_3 , K_4
d.	If $f: A \rightarrow B$ is one to one mapping then prove that $f^{-1}: B \rightarrow A$ will be one to one.	1	[CO2], K_1 , K_2
e.	Find the dual of the boolean expression: $xy(y + y') + x'z$	1	[CO2], K_1 , K_2

SECTION B

Question 2: Attempt any 05 out of the following.

[3X5=15]

a.	Let $R_1 = \{ (1,2), (2,3), (3,4) \}$ and $R_2 = \{ (1,1), (1,2), (2,1), (2,2), (2,3), (3,1), (3,2), (3,3), (3,4) \}$ then find i) $R_1 \cup R_2$ ii) $R_1 \cap R_2$ iii) $R_1 - R_2$ iv) $R_2 - R_1$	3	[CO1], K_3 , K_4
b.	Show that for any sets A & B in set theory $A - (A \cap B) = A - B$. Also draw venn diagram for both.	3	[CO1], K_3 , K_4
c.	Show that the relation xRy iff $(x - y)$ is divisible by 5 is an equivalence relation on the set of integers.	3	[CO1], K_3 , K_4
d.	Define a Lattice. For any a, b, c, d in a lattice $(A, \leq)$ if $a \leq b$ and $c \leq d$ then show that $a \vee c \leq b \vee d$ and $a \wedge c \leq b \wedge d$.	3	[CO1], K_3 , K_4
e.	Let D be the set of all factors of 30 and let $'/'$ be the divisibility relation on L . Then show that $(D, '/')$ is a Lattice.	3	[CO1], K_3 , K_4
f.	Let $f(x) = ax^2 + b$ and $g(x) = cx^2 + d$, where a, b, c, d are constants. Determine for which values of constants $\text{gof}(x) = \text{fog}(x)$.	3	[CO2], K_1 , K_2
g.	Simplify $E = a'b'c' + a'bc' + ab'c' + abc'$.	3	[CO2], K_1 , K_2

SECTION C

Question 3: Attempt any 02 out of the following.

[5X2=10]

a.	Among the first 1000 positive integer (a) Determine the integers which are not divisible by 5, nor by 7, nor by 9 (b) Determine the integers divisible by 5, but not by 7, not by 9	5	[CO1], K_3 , K_4
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b/	Consider the set $D_{50} = \{1, 2, 5, 10, 25, 50\}$ and the relation divides ($\mid$) be a partial ordering relation on D_{50} i) Draw the Hasse diagram of D_{50} with relation divides. ii) Determine all upper bounds of 5 & 10 iii) Determine all lower bounds of 5 & 10 iv) Determine G.L.B of 5 & 10 v) Determine L.U.B of 5 & 10	5	[CO1], K_3 , K_4
c/	Solve $E(x, y, z, t) = \sum(0, 2, 6, 8, 10, 12, 14, 15)$ using K-map	5	[CO2], K_1 , K_2

Bloom's Taxonomy:

K1	K2	K3	K4	K5	K6
Remember	Understand	Apply	Analyze	Evaluate	Create